

Family	Method	Task / suite	Base (%)	ASRR (%)	Δ (pp)	Rescue	Harm
VLA	OpenVLA-OFT	LIBERO-Spatial (500)	92.8 ± 0.0	93.4 ± 0.3	$+0.6 \pm 0.3$	$+8.7 \pm 1.2$	-5.7 ± 2.9
		LIBERO-Object (500)	98.8 ± 0.0	99.1 ± 0.2	$+0.3 \pm 0.2$	$+5.0 \pm 0.0$	-3.3 ± 1.2
		LIBERO-Goal (500)	96.4 ± 0.0	97.3 ± 0.1	$+0.9 \pm 0.1$	$+9.3 \pm 0.6$	-5.0 ± 0.0
		LIBERO-10 (500)	95.2 ± 0.0	96.0 ± 0.0	$+0.8 \pm 0.0$	$+11.0 \pm 0.0$	-7.0 ± 0.0
	Octo*	LIBERO-Spatial (500)	83.1 ± 2.2	85.1 ± 1.6	$+2.0 \pm 1.6$	$+33.0 \pm 8.5$	-23.0 ± 2.6
		LIBERO-Object (500)	60.7 ± 5.0	62.1 ± 4.6	$+1.3 \pm 0.4$	$+22.0 \pm 7.2$	-15.3 ± 5.1
		LIBERO-Goal (500)	87.6 ± 1.0	89.1 ± 1.6	$+1.5 \pm 1.1$	$+38.3 \pm 3.8$	-31.0 ± 3.6
		LIBERO-10 (500)	47.9 ± 2.8	59.5 ± 1.8	$+11.6 \pm 1.0$	$+104.7 \pm 7.0$	-46.7 ± 6.7
	$\pi_{0.5}$ *	LIBERO-Spatial (500)	84.7 ± 0.3	93.6 ± 2.2	$+8.9 \pm 2.4$	$+62.3 \pm 10.2$	-17.7 ± 2.9
		LIBERO-Object (500)	88.2 ± 0.7	99.1 ± 0.6	$+10.9 \pm 0.6$	$+58.3 \pm 2.5$	-3.7 ± 2.3
		LIBERO-Goal (500)	71.3 ± 3.0	82.1 ± 2.5	$+10.9 \pm 2.8$	$+74.0 \pm 9.0$	-19.7 ± 5.1
		LIBERO-10 (500)	68.4 ± 0.3	84.1 ± 1.7	$+15.7 \pm 1.9$	$+110.3 \pm 7.2$	-32.0 ± 4.4
	SmolVLA	LIBERO-Spatial (500)	67.4 ± 1.2	68.5 ± 0.5	$+1.1 \pm 1.0$	$+98.0 \pm 3.6$	-92.3 ± 1.5
		LIBERO-Object (500)	79.6 ± 1.3	80.8 ± 0.9	$+1.2 \pm 1.1$	$+56.7 \pm 7.6$	-50.7 ± 2.1
		LIBERO-Goal (500)	73.3 ± 1.0	76.1 ± 1.3	$+2.8 \pm 2.1$	$+62.0 \pm 9.5$	-48.0 ± 4.4
		LIBERO-10 (500)	33.4 ± 0.6	36.9 ± 2.3	$+3.5 \pm 2.7$	$+83.7 \pm 9.9$	-66.3 ± 3.8

Table 1: Three-round VLA ASRR evaluations on LIBERO. Base and Base+ASRR columns are success percentages reported as mean $\pm$ sample standard deviation over three rounds; each round evaluates 500 paired rollouts per suite. Δ is the Base+ASRR improvement over Base in percentage points, and rescue/harm count paired rollouts where ASRR turns a Base failure into success or a Base success into failure. OpenVLA-OFT and SmolVLA use fully fine-tuned LIBERO policies, whereas Octo and $\pi_{0.5}$ use undertrained LoRA-finetuned policies. ASRR gives positive gains for all VLA policies and is especially effective for undertrained models.